

Xuan Huang

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Personal Website

Educational Background

University of Electronic Science and Technology of China

Sept. 2022 - Jun. 2026 (expected)

B.Eng in Communication Engineering, SICE

- Rank – 4/189 (top 2.1%) Average Score – 94.2/100 GPA – 3.93/4
- Major courses: Digital Signal Processing (99), Probability & Statistics (96), Information Theory (93)
- Language: CET-4: 664 CET-6: 625

Publications

- [C1] X. Huang, Y. Zhang, X. Zhang, X. Zhang, and H. Tang, “Voice Recognition System for Speech-to-Text and GPT Communication Powered by Organic Photovoltaic,” accepted by *ICDT*, 2025.
- [C2] X. Wei, W. Mei, X. Huang, Z. Chen, and B. Ning, “Mechanical Power Modeling and Energy Efficiency Maximization for Movable Antenna Systems,” accepted by *IEEE Globecom*, 2025[PDF].

Honors and Awards

- 2023-2024 National Scholarship (Top 1.5%) for Undergraduates
- 2022-2023 & 2023-2024 UESTC Outstanding Student Scholarship
- 2024 Huawei Cup National IoT Design Competition National First Prize
- 2024 Contemporary Undergraduate Mathematical Contest in Modeling National Second Prize
- 2024 National English Competition for College Students (NECCS) National Second Prize
- 2024 Mathematical Contest in Modeling (MCM) Honorable Mention

Research Experiences

Self-powered Multimodal Emotion Recognition System

Dec. 2023 – Aug. 2024

- Multimodal Fusion Model:** Built a dual-modal model with modality-specific/invariant feature fusion for speech and wavelet-based EEG preprocessing for ViT input.
- IoT System Development:** Developed full-stack system with ESP32-based EEG acquisition, Huawei Cloud for IoT data storage, Alibaba Cloud for backend model deployment, and WeChat Mini Program frontend.
- WeChat Mini Program:** Designed visualization interface supporting emotion monitoring and mood diary features.
- Published a **First-author** paper which is accepted by *ICDT 2025* (see Section “Publications”).
- National First Prize** in Huawei IoT Design Competition (Top 3.6%).

Movable Antenna-Assisted Wireless Communication System

Dec. 2024 – Mar. 2025

- Mathematical Modeling Support:** Participated in the derivation of the stepper motor power model and the global EE optimization problem; analyzed the impact of movement speed on motor torque and system energy efficiency.
- Literature Review:** Reviewed related work on movable antenna systems and contributed to the technical background.
- Published a **Second student author** paper which is accepted by *IEEE Globecom 2025* (see Section “Publications”).

Tunable Structural Color Based on Multilayer Films

Dec. 2023 – Aug. 2024

- Transfer Matrix Modeling:** Derived electromagnetic field transfer matrices for Fabry–Pérot multilayer cavities; established global transmission model using boundary conditions and tangential continuity equations.
- Reflectance Simulation and Optimization:** Developed MATLAB code for dynamic reflectance prediction; optimized layer thicknesses and dielectric constants to simulate tunable color performance.

Skills and Services

- Technical skills:** Experienced in **Python**, **C/C++** and **Matlab**. Available for **JavaScript** and **Verilog**.
- Social services:** Certified Peer Counselor at UESTC, Class Study Committee Member (“Excellent Class”).